SAMPLE

# Famous Astronomers: Fritz Zwicky

Dark Matter Pioneer

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## Introduction

Fritz Zwicky occupies a singular seat in the gallery of twentieth-century astronomy, the profession's most productive heretic, the Swiss outsider who looked at the universe, found it mostly missing, and spent fifty years saying so in print. The man's name belongs to four discoveries that each deserved a Nobel and received none: the supernova understood as the death of a massive star, the neutron star predicted two years before anyone could dream of detecting one, dark matter deduced from the misbehaving galaxies a third of a century before the field conceded it existed, and the compact galaxies whose light harbored the first hints of some of the deepest engines in the sky. The volume's subject being the archetypal difficult genius, the reader should be warned at the threshold: the man was as abrasive as he was right, and the history is partly the story of a field that took its best ideas from someone it could not stomach.

The chapters ahead follow the library's standard architecture, the sixteen rooms the sibling biographies share: the discovery of the man and his arc, the character itself, the science he made, the instruments he built and used, the features of his career that every account must include, the contexts of his time, the modern verdict on his ideas, the research programs he launched, the discoveries ledger, the mysteries he left, the open ends of his research programs, the trivia shelf, and the significance verdict that closes the book. A veteran of the sibling volumes will recognize the furniture; a newcomer may enter by any door and lose nothing; and the subject being one of the series' richest.

The volume's method is the series' own: no clutter, no citation scaffolding, no borrowed encyclopedia furniture, only the continuous catalog voice telling the whole story. The biography being the spine, the science being the muscle, and the personality being the friction that shaped both, the book aiming to leave the reader with the man whole: the mountaineer, the jet engineer, the catalog maker, the teacher, and the prophet whom the century eventually, reluctantly, vindicated.

The volume's warnings being given, the volume's promises should follow: the reader being promised the whole man rather than the cartoon, the science being explained rather than merely reported, the instruments being shown at work, and the gossip being kept in its honest proportion, the book being neither the prosecution's brief nor the family's album. The series' standard being the biography with the arithmetic kept in, this volume carrying some of the heaviest arithmetic in the library, and the reader being invited to check every number the man checked, the invitation being the book's own method and the man's own creed.

The volume's factual spine being the standard record: the birth in Varna in eighteen ninety-eight, the doctorate at Zurich in nineteen twenty-two, the arrival at Caltech in nineteen twenty-five, the dark matter paper in nineteen thirty-three, the supernova and neutron star paper with Baade in nineteen thirty-four, the eighteen-inch Schmidt at Palomar from nineteen thirty-six, the rocket and jet years in the forties, the Sky Survey advocacy in the late forties and fifties, the compact catalogs and the cluster atlases across the fifties and sixties, the National Medal of Science in nineteen seventy-two, and the death in February of nineteen seventy-four, the spine being the dates the chapters hang their stories on, and the reader being invited to check them against the archive the biography cites in its final pages.

## Chapter 1: The Discovery of Fritz Zwicky

The man's own beginning was itself a crossing of borders, the astronomer having been born in Varna on the Bulgarian coast of the Black Sea in eighteen ninety-eight, the son of a Swiss merchant family, the father's grain business having carried the name to the far shore of Europe. The household's being Swiss in a Bulgarian port being the first of the dislocations that marked the life: the boy's German-speaking home inside a Slavic-speaking city, the family's return to Switzerland when the boy was six, the father's early death leaving the boy to be raised in the village of Mollis in the canton of Glarus, the grandmother's strict household being the discipline behind the later iron.

The education being Swiss and formidable: the Gymnasium years in Zurich, the mathematics and physics studies at the Eidgenossische Technische Hochschule, the doctorate in nineteen twenty-two on the mathematics of surfaces, the thesis being pure mathematics rather than astronomy, the young doctor's having arrived at the stars by the side door of theory. The early work's being on crystallography and ionic structures, the solid-state years being the apprenticeship that later showed: the physicist who had computed lattices being the same one who would compute the density of the invisible, the training in exact structure being the habit the cosmos would

eventually answer to.

The crossing to America being the decisive discovery: the Rockefeller fellowship's having brought the twenty-four-year-old to Caltech in nineteen twenty-five, the Pasadena campus being then barely a decade old, the physics department's being a startup with ambitions, the young Swiss's having been invited to work on quantum theory with the great men and having stayed instead for fifty-one years, the visitors' visa having become a life. The discovery of the man being thus an American discovery too: the European training meeting the California money, the resulting alloy being the astronomer the century could not classify.

The tenure at Caltech being contemporaneous with the arrival of the great telescope era: the Mount Wilson hundred-inch being the world's largest while the young theorist was computing, the two-hundred-inch being planned during his middle years, the Palomar survey being partly his own doing, the man's having been at the hinge of the telescope century, the theorist who became one of the great observers and catalogers, the discovery chapter's arc being the transformation: from the mathematician of surfaces to the astronomer of the missing, the border-crossing boy having crossed one border too many for comfort and having ended at the edge of the visible itself.

The war years being the discovery of the engineer: the rocket and jet work with the Caltech group that became the Jet Propulsion Laboratory, the Aerojet company's having been co-founded with the rocket men of Pasadena, the propulsion work being classified and public by turns, the astronomer's having spent the war years building the machines that would carry later astronomers' instruments, the man's having been an engineer of escape velocities in the day job and an engineer of invisible masses at night, the two careers being one habit: the computation of trajectories and the reading of the counterintuitive.

The postwar decades being the discovery of the cataloger: the eighteen-inch Schmidt telescope at Palomar being the instrument of the supernova hunts, the thousands of photographic plates being the archive, the compact galaxies being the catch, the six-volume catalogs being the monument. The professorship having been earned decades earlier, the National Medal of Science having landed only in seventy-two, the honor's having arrived when the vindication was nearly complete, the discoverer's having been discovered by his own field four decades after the first heresies, the chapter's closing being the long arc: the boy from Varna having become the old man of Pasadena whose every early insult had matured into a citation.

The discovery chronology deserves a fuller record, the man's having been visible early. The first papers of the twenties being the solid-state work, the reputation's having been made in the quantum-mechanics seminars of Zurich and Pasadena, the young theorist's having been taken seriously by the best before thirty; the turning point being the arrival at the astrophysical questions around nineteen twenty-eight, the man's having begun asking what the new physics did to the old stars, the neutron's arriving in nineteen thirty-two being the trigger, the dark matter and the neutron star being the two answers the following twenty-four months produced, the decade's output being the foundation of the modern field, and the man's having been thirty-five when he finished the century's homework early.

The network being part of the discovery: the man's having been surrounded by the era's best, Millikan being the institute's president and the champion of the hire, Baade being the observational partner, Hubble being the neighbor at the mountain, Oppenheimer and the rocket men being the colleagues of the war years. The environment's being the multiplier of the talent, the man's heresies being argued with the elite rather than the fringe, the discovery of the man being partly the discovery of the room he walked into, the room being the best in the world, and the man's having treated it as his personal sparring gym, the biography being partly the roster of the era and the man's being the era's most combative resident.

The late life being the discovery's epilogue: the man's having spent the sixties and early seventies writing the catalogs and the morphological books, the vindication's having arrived in installments, the quasars' and the pulsars' having knocked on the door of the old claims, the National Medal's arriving in seventy-two, the death's coming in February of nineteen seventy-four, the man's having died before the dark matter's full return but after the neutron star's, the partial apology being the chronology's irony, the man's last years being spent watching his wildest prediction become textbook while his largest computation still waited for the committee, the epilogue being the race the man lost by two years and won by eternity.

The discovery of the man's methods being the discovery of the man: the era's physics having been in ferment, the man's approach having been the application of the new mechanics to the stubborn old data, the pattern repeating across the career, the dark matter being the virial applied to clusters, the neutron star being the quantum applied to

the collapsing core, the compact galaxies being the diligence applied to the plates, the morphological box being the same impulse turned on invention itself, the man's having been one method wearing four costumes, and the method being the counting of what others waved at, the discovery chapter having found the man early and the method being the signature the whole biography carries.

Consider the timing of the arrival once more, because the coincidence deserves its own paragraph. In nineteen twenty-five the quantum mechanics was two years old, the expanding universe was four years in the future, the neutron was seven years away, and the largest telescope in the world was being dedicated on the mountain above Pasadena. A twenty-seven-year-old theorist stepping off the train that year was stepping onto a stage where every prop of the coming century was being unpacked at once. Zwicky spent the next decade assembling those props into the two papers this biography keeps returning to, and the decade's other residents being the best supporting cast the century ever assembled, the young Swiss's having been the right immigrant at the right station at the single right moment, and the discovery chapter being the record of that alignment, the alignment having been luck and the use of it having been the man's own.

One discovery-era story deserves retelling at full length, the tale being the man's own favorite. The visiting astronomer being shown the mountain, the young Swiss's having been bored by the social climbing of the profession, the man's having proposed that the mountain's time be spent on the mountain's work, the anecdote being the era's compressed biography: the theorist who believed the glass belonged to the counters and the catalogers rather than the ceremonialists. Whether the specific retelling is embroidered being the biographer's caution, the story's surviving because it is true in the way legends are true, the shape being accurate even where the seam shows, the discovery chapter keeping the legend with the annotation, and the reader being given the man the way the era received him: partly in focus, partly in myth, and wholly in motion.